Spatial Field Filtering and SGS Structure Learning

A three-location Spatial Durbin Model

with linear Gaussian DAG-SEM examples

Working thesis presentation ▪ November 2026

One question, two inputs to SGS

Original outcomes

$$(y_1, y_2, y_3)$$

Spatial dependence remains in the observed outcomes.

Field-filtered outcomes

$$(z_1, z_2, z_3)$$

The *realized* SDM field has been subtracted.

Question: does the graph found by SGS change? It may find a different skeleton (the set of undirected edges), a different collider pattern, or no edges.

Throughout: nodes 1, 2, and 3 are geographical locations, not three types of outcome.

The whole argument in four steps

1. **Original outcomes mix disturbances.** Through Wy, the outcome at location i contains the disturbances of the locations in its equation, and of their neighbours.
2. **Exact filtering removes that channel.** Each filtered value keeps only its own disturbance ε_i.
3. **Dependence needs a shared disturbance.** Two Gaussian outcomes are marginally dependent only if they contain a common disturbance. Conditioning on a third outcome can remove this dependence (chain, fork) or create it (collider).
4. **Consequence.** Filtered data: no shared disturbance, so SGS finds no edges. Original data: the sharing pattern, fixed by W, decides the SGS graph.

The following slides derive each step.

What the analysis does—and does not—identify

1. **Algebra:** derive the joint distribution of each SGS input.
2. **Independence:** compute all possible three-node Gaussian SGS checks.
3. **Graph:** compare the resulting undirected edges and compelled colliders.

Boundary of the result

A change in an SGS graph does *not* prove that the filtered graph is the true causal structure.

These are population calculations. Estimation and repeated sampling are separate questions.

The only spatial model used in this presentation

$$y = \rho W y + \beta x + \theta W x + \varepsilon$$

Symbol	Meaning
$y \in \mathbb{R}^3$	three outcomes
$x \in \mathbb{R}^3$	one covariate at each location
$W \in \mathbb{R}^{3\times 3}$	known spatial weights
ρ	lagged-outcome coefficient

Symbol	Meaning
β	local-covariate coefficient
θ	spatial-covariate coefficient
$\varepsilon \in \mathbb{R}^3$	structural disturbances
w_{ij}	weight of location j in row i

The model is scalar: one outcome variable, measured at three locations.

Explicitly, one equation per location

$$
\begin{aligned}
y_1 &= \rho(w_{11}y_1 + w_{12}y_2 + w_{13}y_3) + \beta x_1 \\
&\quad + \theta(w_{11}x_1 + w_{12}x_2 + w_{13}x_3) + \varepsilon_1, \\
y_2 &= \rho(w_{21}y_1 + w_{22}y_2 + w_{23}y_3) + \beta x_2 \\
&\quad + \theta(w_{21}x_1 + w_{22}x_2 + w_{23}x_3) + \varepsilon_2, \\
y_3 &= \rho(w_{31}y_1 + w_{32}y_2 + w_{33}y_3) + \beta x_3 \\
&\quad + \theta(w_{31}x_1 + w_{32}x_2 + w_{33}x_3) + \varepsilon_3.
\end{aligned}
$$

Convention: $w_{ij} \neq 0$ means location j enters location i's outcome equation.

Assumptions for the exact comparison

$$\boxed{\varepsilon \mid x \sim N(0, \sigma^2 I), \quad \sigma^2 > 0, \quad I - \rho W \text{ invertible}}$$

- The spatial weights W and coefficients ρ, β, θ are fixed.
- The covariate vector x is fixed or explicitly conditioned upon.
- The disturbances $\varepsilon_1, \varepsilon_2, \varepsilon_3$ are mutually independent Gaussian variables, each with variance σ^2.
- SGS is a **population oracle**: every independence test gives the correct answer (no sampling error).

We use $0 < \rho < 1$ for the numerical patterns in the worked examples.

Original = local + realized field + error

$$\underbrace{y}_{\text{original}} = \underbrace{\beta x}_{\text{local}} + \underbrace{(\rho W y + \theta W x)}_{\text{realized field}} + \underbrace{\varepsilon}_{\text{error}}.$$

The realized field is the part that comes from the neighbours. It uses the observed y, so it contains this draw's disturbances:

$$\boxed{f = \rho W y + \theta W x.}$$

Exact field filtering subtracts it:

$$\begin{aligned} z &= y - f && \text{definition of the filter} \\ &= (\rho W y + \beta x + \theta W x + \varepsilon) - (\rho W y + \theta W x) && \text{insert the SDM for } y \text{ and the definition of } f \\ &= \boxed{\beta x + \varepsilon} && \rho W y \text{ and } \theta W x \text{ cancel} \end{aligned}$$

The disturbance ε is still present after field filtering.

The two SGS inputs: a crucial distinction

Original observations	Exactly filtered observations
$y = \rho W y + \beta x + \theta W x + \varepsilon$	$z = y - (\rho W y + \theta W x) = \beta x + \varepsilon$
Potentially spatially dependent.	Independent across locations when x is fixed, because $\varepsilon_1, \varepsilon_2, \varepsilon_3$ are independent.

Do **not** confuse $z = \beta x + \varepsilon$ with βx alone. When x is fixed, βx is the same in every draw: its variance is zero, so SGS has nothing to test.

Solve the simultaneous SDM: reduced form

Question: y appears on both sides of the SDM. What is y in terms of x and ε only?

$$y = \rho W y + \beta x + \theta W x + \varepsilon \qquad \text{the SDM}$$

$$y - \rho W y = \beta x + \theta W x + \varepsilon \qquad \text{subtract } \rho W y \text{ from both sides}$$

$$(I - \rho W) y = (\beta I + \theta W) x + \varepsilon \qquad \text{write } y = Iy,\ x = Ix, \text{ then factor}$$

$$\boxed{y = (I - \rho W)^{-1}(\beta I + \theta W) x + (I - \rho W)^{-1} \varepsilon} \qquad \text{multiply on the left by } (I - \rho W)^{-1}$$

Reduced form: the right-hand side contains x and ε, but no y.

The inverse $(I - \rho W)^{-1}$ exists by assumption.

After centering, only the disturbances vary

$$\mathrm{E}[y \mid x] = (I - \rho W)^{-1}(\beta I + \theta W)x \qquad \text{mean of the reduced form; } \mathrm{E}[\varepsilon \mid x] = 0$$

$$\widetilde{y} := y - \mathrm{E}[y \mid x] = (I - \rho W)^{-1}\varepsilon \qquad \text{subtract the mean; the } x \text{ term cancels}$$

Read $(I - \rho W)^{-1}$ as a mixing table

Row i gives how much of $\varepsilon_1, \varepsilon_2, \varepsilon_3$ is inside the centered outcome $\widetilde{y}_i$.

Example (chain $1 \to 2 \to 3$, Example 1): row 3 is $(\rho^2, \rho, 1)$, so $\widetilde{y}_3 = \rho^2\varepsilon_1 + \rho\varepsilon_2 + \varepsilon_3$.

β and θ appear only in the mean. With x fixed, the spatial dependence of $\widetilde{y}$ is set by W, ρ, and σ^2 alone.

One rule for every covariance: shared disturbances

Question: when are two centered outcomes correlated? Write each as a mix of the disturbances:

$$\widetilde{y}_i = a_1\varepsilon_1 + a_2\varepsilon_2 + a_3\varepsilon_3, \qquad \widetilde{y}_j = b_1\varepsilon_1 + b_2\varepsilon_2 + b_3\varepsilon_3.$$

Expand the covariance. Cross terms vanish because $\mathrm{Cov}(\varepsilon_k, \varepsilon_l) = 0$ for $k \neq l$, and $\mathrm{Var}(\varepsilon_k) = \sigma^2$:

$$\mathrm{Cov}(\widetilde{y}_i, \widetilde{y}_j) = \sigma^2 \left(a_1 b_1 + a_2 b_2 + a_3 b_3\right).$$

Shared disturbance (chain):
$\widetilde{y}_1 = \varepsilon_1, \;\; \widetilde{y}_3 = \rho^2\varepsilon_1 + \rho\varepsilon_2 + \varepsilon_3$
$\mathrm{Cov} = \sigma^2(1 \cdot \rho^2) = \rho^2\sigma^2 \neq 0.$

No shared disturbance (filtered):
$z_1 - \beta x_1 = \varepsilon_1, \;\; z_3 - \beta x_3 = \varepsilon_3$
$\mathrm{Cov} = \sigma^2(1 \cdot 0 + 0 \cdot 0 + 0 \cdot 1) = 0.$

Two outcomes can be correlated only if they contain at least one common disturbance.

Original and filtered covariance matrices

The same rule, for all pairs at once. Let $\Sigma := \mathrm{Cov}(y \mid x)$:

$$
\begin{aligned}
\Sigma &= \mathrm{E}[\widetilde{y}\,\widetilde{y}^{\mathsf{T}} \mid x] && \text{definition; } \widetilde{y} = y - \mathrm{E}[y \mid x] \\
&= \mathrm{E}[(I-\rho W)^{-1}\varepsilon\varepsilon^{\mathsf{T}}(I-\rho W)^{-\mathsf{T}} \mid x] && \text{insert } \widetilde{y} = (I-\rho W)^{-1}\varepsilon\text{; use } (AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}} \\
&= (I-\rho W)^{-1}\ \mathrm{E}[\varepsilon\varepsilon^{\mathsf{T}} \mid x]\,(I-\rho W)^{-\mathsf{T}} && \text{the matrices are fixed} \\
&= \boxed{\sigma^2 (I-\rho W)^{-1}(I-\rho W)^{-\mathsf{T}}} && \mathrm{E}[\varepsilon\varepsilon^{\mathsf{T}} \mid x] = \sigma^2 I
\end{aligned}
$$

For the filtered outcomes, $z - \mathrm{E}[z \mid x] = (\beta x + \varepsilon) - \beta x = \varepsilon$. Thus

$$\boxed{\mathrm{Cov}(z \mid x) = \sigma^2 I.}$$

Here $A^{-\mathsf{T}}$ means $(A^{-1})^{\mathsf{T}}$. Each filtered value contains only its own disturbance, so every off-diagonal covariance is zero.

Three nodes mean exactly six possible SGS tests

Pair	Marginal independence	Conditional independence
$(1,2)$	$y_1 \perp\!\!\!\perp y_2 \mid x$	$y_1 \perp\!\!\!\perp y_2 \mid (y_3, x)$
$(1,3)$	$y_1 \perp\!\!\!\perp y_3 \mid x$	$y_1 \perp\!\!\!\perp y_3 \mid (y_2, x)$
$(2,3)$	$y_2 \perp\!\!\!\perp y_3 \mid x$	$y_2 \perp\!\!\!\perp y_3 \mid (y_1, x)$

1. Start with all three undirected edges.
2. Remove an edge if *either* independence check finds a separating set (a conditioning set that makes the pair independent).
3. Orient $1 \to 2 \leftarrow 3$ only when the path endpoints were separated *without* conditioning on node 2.

Gaussian test, part 1: marginal independence

The original triple is jointly Gaussian. Therefore

$$\boxed{y_i \perp\!\!\!\perp y_j \,|\, x \quad \Longleftrightarrow \quad \Sigma_{ij} = 0.}$$

Example: if $\Sigma_{13} = 0$, SGS removes edge $1 - 3$ with the empty conditioning set, that is, without conditioning on node 2.

Why joint Gaussianity is needed

Without it, zero covariance does not imply independence. Example: $u \sim N(0, 1)$ and $v = u^2$. Then $\mathrm{Cov}(u, v) = \mathrm{E}[u^3] = 0$, but v is a function of u, so the two are dependent. The pair (u, v) is not jointly Gaussian.

Gaussian test, part 2: remove what y_k predicts

Question: once y_k is known, is any association left between y_i and y_j?

On this slide and the next, $\widetilde{y}_i, \widetilde{y}_j, \widetilde{y}_k$ are centered outcomes, $\widetilde{y} = y - \mathrm{E}[y \mid x]$; x stays fixed.

Step 1. Remove from $\widetilde{y}_i$ and $\widetilde{y}_j$ the part that $\widetilde{y}_k$ predicts (regression coefficients Σ_{ik}/Σ_{kk} and Σ_{jk}/Σ_{kk}):

$$r_i = \widetilde{y}_i - \frac{\Sigma_{ik}}{\Sigma_{kk}}\widetilde{y}_k, \qquad r_j = \widetilde{y}_j - \frac{\Sigma_{jk}}{\Sigma_{kk}}\widetilde{y}_k.$$

Step 2. The remainder r_i no longer moves with $\widetilde{y}_k$:

$$\mathrm{Cov}(r_i, \widetilde{y}_k) = \Sigma_{ik} - \frac{\Sigma_{ik}}{\Sigma_{kk}}\Sigma_{kk} = 0.$$

Step 3. Gaussian and zero covariance imply that r_i, r_j are independent of $\widetilde{y}_k$. Given y_k, the subtracted terms are fixed numbers, so only r_i, r_j still vary:

$$\boxed{\mathrm{Cov}(y_i, y_j \mid y_k, x) = \mathrm{Cov}(r_i, r_j).}$$

Gaussian test, part 2: the conditional covariance formula

Expand $\mathrm{Cov}(r_i, r_j)$ with $r_i = \widetilde{y}_i - \frac{\Sigma_{ik}}{\Sigma_{kk}}\widetilde{y}_k$ and $r_j = \widetilde{y}_j - \frac{\Sigma_{jk}}{\Sigma_{kk}}\widetilde{y}_k$:

$$\mathrm{Cov}(r_i, r_j) = \Sigma_{ij} - \frac{\Sigma_{jk}}{\Sigma_{kk}}\Sigma_{ik} - \frac{\Sigma_{ik}}{\Sigma_{kk}}\Sigma_{jk} + \frac{\Sigma_{ik}\Sigma_{jk}}{\Sigma_{kk}^2}\Sigma_{kk} \qquad \text{expand the four products}$$

$$= \Sigma_{ij} - 2\,\frac{\Sigma_{ik}\Sigma_{jk}}{\Sigma_{kk}} + \frac{\Sigma_{ik}\Sigma_{jk}}{\Sigma_{kk}} \qquad \text{middle terms are equal; cancel } \Sigma_{kk}$$

$$= \boxed{\Sigma_{ij} - \frac{\Sigma_{ik}\Sigma_{jk}}{\Sigma_{kk}}} \qquad \text{combine}$$

How to read it: total covariance minus the covariance that goes through y_k

Removed: chain $1 \to 2 \to 3$. Total $\rho^2\sigma^2$; through 2, also $\rho^2\sigma^2$. Nothing is left (Example 1).

Created: collider $1 \to 2 \leftarrow 3$. Total 0; through 2, $\rho^2\sigma^2/(1+2\rho^2) > 0$. The remainder is negative (Example 4).

A shortcut: the Gaussian precision matrix

Let $\Omega = \Sigma^{-1}$. Start from $\Sigma = \sigma^2 (I - \rho W)^{-1} (I - \rho W)^{-\mathsf{T}}$:

$$\Omega = \frac{1}{\sigma^2} (I - \rho W)^{\mathsf{T}} (I - \rho W) \qquad (AB)^{-1} = B^{-1} A^{-1}\text{: invert the factors in reverse order}$$

$$= \frac{1}{\sigma^2} \left[I - \rho W - \rho W^{\mathsf{T}} + \rho^2 W^{\mathsf{T}} W \right] \qquad \text{multiply out}$$

$$\boxed{\Omega_{ij} = \frac{-\rho(w_{ij} + w_{ji}) + \rho^2 \sum_{k=1}^{3} w_{ki} w_{kj}}{\sigma^2}, \quad i \neq j} \qquad \text{uses } (W^{\mathsf{T}} W)_{ij} = \sum_k w_{ki} w_{kj}$$

The test

$\Omega_{ij} = 0$ if and only if $y_i \perp\!\!\!\perp y_j \,|\, (y_k, x)$, with k the third node. Reason:
$\Omega_{ij} = -\Sigma_{kk} \operatorname{Cov}(y_i, y_j \,|\, y_k, x) / \det \Sigma$ (appendix).

How to read it. $\Omega_{ij} \neq 0$ when i and j are directly linked (w_{ij} or $w_{ji} \neq 0$), or when both enter the equation of the same location k ($w_{ki} w_{kj} \neq 0$), unless the two terms cancel exactly.

The SDM as a linear Gaussian DAG-SEM

$$y_i = \rho \sum_{j \in \mathrm{Pa}(i)} w_{ij} y_j + \beta x_i + \theta \sum_{j=1}^{3} w_{ij} x_j + \varepsilon_i$$

- $\mathrm{Pa}(i)$, the parents of i: the locations whose outcomes appear in i's equation.
- Nonzero w_{ij} means the arrow $j \to i$.
- An acyclic W (no directed cycle, zero diagonal) makes this a linear Gaussian DAG-SEM: a structural equation model on a directed acyclic graph.



Acyclic example:
$\mathrm{Pa}(1) = \varnothing$
$\mathrm{Pa}(2) = \{1\}$
$\mathrm{Pa}(3) = \{2\}$

This is the same scalar SDM, not a new multivariate model.

Each node is Gaussian given its parents

Isolate the disturbance in the equation of location i:

$$\varepsilon_i = y_i - \rho \sum_{j \in \mathrm{Pa}(i)} w_{ij} y_j - \beta x_i - \theta \sum_{j=1}^{3} w_{ij} x_j .$$

Fix the parent outcomes and x. Then only $\varepsilon_i \sim N(0, \sigma^2)$ is random, and adding fixed numbers to it moves only its mean:

$$\boxed{y_i \mid (y_{\mathrm{Pa}(i)}, x) \sim N\left(\rho \sum_{j \in \mathrm{Pa}(i)} w_{ij} y_j + \beta x_i + \theta \sum_{j} w_{ij} x_j,\ \sigma^2 \right).}$$

Acyclic: the nodes can be ordered so that every parent comes before its child. Each ε_i is new at node i, so it is independent of the earlier nodes.

The joint density is a product over nodes

$$\boxed{p(y_1, y_2, y_3 \mid x) = \prod_{i=1}^{3} p(y_i \mid y_{\mathrm{Pa}(i)}, x).}$$

Concrete case, the chain $1 \to 2 \to 3$ ($\mathrm{Pa}(1) = \varnothing$, $\mathrm{Pa}(2) = \{1\}$, $\mathrm{Pa}(3) = \{2\}$):

$$\begin{aligned} p(y_1, y_2, y_3 \mid x) &= p(y_1 \mid x)\, p(y_2 \mid y_1, x)\, p(y_3 \mid y_1, y_2, x) && \text{chain rule of probability (always true)} \\ &= p(y_1 \mid x)\, p(y_2 \mid y_1, x)\, p(y_3 \mid y_2, x) && y_3 \text{ is made from } y_2,\ x,\ \varepsilon_3 \text{ only} \end{aligned}$$

Once y_2 and x are given, y_1 does not change the distribution of y_3:

$$y_1 \perp\!\!\!\perp y_3 \mid (y_2, x).$$

This is the independence that SGS finds in Example 1.

D-separation: noncollider versus collider

d-separation is the rule that reads from the arrows alone which conditional independences every distribution on the DAG must have.

Noncollider path



$y_1 \not\perp\!\!\!\perp y_3 \mid x, \qquad \boxed{y_1 \perp\!\!\!\perp y_3 \mid (y_2, x)}$

$\{2\}$ separates 1 and 3.
Conditioning on 2 blocks the path.

Collider

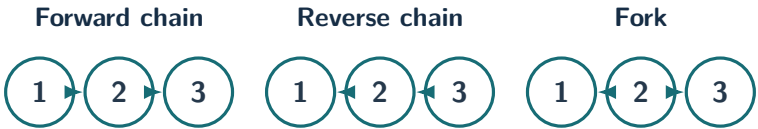


$\boxed{y_1 \perp\!\!\!\perp y_3 \mid x}, \qquad y_1 \not\perp\!\!\!\perp y_3 \mid (y_2, x)$

$\varnothing$ separates 1 and 3.
Conditioning on the common effect 2 opens the path.

SGS orients a collider *only* when the middle node was not in the separating set.

Acyclic does not mean uniquely orientable



All three imply the same observed independence pattern:

$$y_1 \perp\!\!\!\perp y_3 \,|\, (y_2, x), \qquad y_1 \not\perp\!\!\!\perp y_3 \,|\, x.$$

Population SGS returns the same unoriented path $1 - 2 - 3$.

DAGs that imply the same conditional independences are called Markov equivalent.

Example 1: forward chain, $1 \to 2 \to 3$

$$W = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$



Centering the outcomes ($\widetilde{y} = y - \mathrm{E}[y \mid x]$) gives:

$$\widetilde{y}_1 = \varepsilon_1 \qquad \text{no parents}$$
$$\widetilde{y}_2 = \rho\widetilde{y}_1 + \varepsilon_2 = \rho\varepsilon_1 + \varepsilon_2 \qquad \text{insert } \widetilde{y}_1$$
$$\widetilde{y}_3 = \rho\widetilde{y}_2 + \varepsilon_3 = \rho^2\varepsilon_1 + \rho\varepsilon_2 + \varepsilon_3 \qquad \text{insert } \widetilde{y}_2$$

Before computing: $\widetilde{y}_1$ and $\widetilde{y}_3$ share ε_1, so they are marginally dependent. Given $\widetilde{y}_2$, the rest of $\widetilde{y}_3$ is ε_3, which is independent of $\widetilde{y}_1 = \varepsilon_1$. So we expect $y_1 \perp\!\!\!\perp y_3 \,|\, (y_2, x)$.

The covariate and Durbin terms vanish from the centered equations because x is fixed.

Example 1: compute covariance and delete $1-3$

Shared-disturbance rule, with $\widetilde{y}_1 = \varepsilon_1$, $\widetilde{y}_2 = \rho\varepsilon_1 + \varepsilon_2$, $\widetilde{y}_3 = \rho^2\varepsilon_1 + \rho\varepsilon_2 + \varepsilon_3$:

$$\Sigma_{13} = \rho^2\sigma^2, \quad \Sigma_{12} = \rho\sigma^2, \quad \Sigma_{23} = (\rho + \rho^3)\sigma^2, \quad \Sigma_{22} = (1+\rho^2)\sigma^2.$$

$$\begin{aligned}
\mathrm{Cov}(y_1, y_3 \mid y_2, x) &= \Sigma_{13} - \frac{\Sigma_{12}\Sigma_{23}}{\Sigma_{22}} && \text{conditional covariance formula} \\
&= \rho^2\sigma^2 - \frac{(\rho\sigma^2)(\rho+\rho^3)\sigma^2}{(1+\rho^2)\sigma^2} && \text{insert the four entries} \\
&= \rho^2\sigma^2 - \frac{\rho^2(1+\rho^2)\sigma^2}{1+\rho^2} = \rho^2\sigma^2 - \rho^2\sigma^2 = \boxed{0} && \rho(\rho+\rho^3) = \rho^2(1+\rho^2);\ \text{cancel}
\end{aligned}$$

SGS on the originals

Remove $1-3$. Keep $1-2$ and $2-3$: directly linked, $\Omega_{12} = \Omega_{23} = -\rho/\sigma^2 \neq 0$. The separating set $\{2\}$ contains the middle node: *no* collider.

Example 2: reverse chain, $1 \leftarrow 2 \leftarrow 3$

$$W = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$



$$\widetilde{y}_3 = \varepsilon_3 \qquad \text{no parents}$$
$$\widetilde{y}_2 = \rho\widetilde{y}_3 + \varepsilon_2 = \rho\varepsilon_3 + \varepsilon_2 \qquad \text{insert } \widetilde{y}_3$$
$$\widetilde{y}_1 = \rho\widetilde{y}_2 + \varepsilon_1 = \varepsilon_1 + \rho\varepsilon_2 + \rho^2\varepsilon_3 \qquad \text{insert } \widetilde{y}_2$$

$$\Sigma_{13} = \rho^2\sigma^2, \quad \Sigma_{12} = (\rho + \rho^3)\sigma^2, \quad \Sigma_{23} = \rho\sigma^2, \quad \Sigma_{22} = (1 + \rho^2)\sigma^2.$$

$$\mathrm{Cov}(y_1, y_3 \mid y_2, x) = \Sigma_{13} - \frac{\Sigma_{12}\Sigma_{23}}{\Sigma_{22}} = \rho^2\sigma^2 - \frac{(\rho + \rho^3)\sigma^2 \cdot \rho\sigma^2}{(1 + \rho^2)\sigma^2} = 0.$$

Given $\widetilde{y}_2$, the rest of $\widetilde{y}_1$ is ε_1, which is independent of $\widetilde{y}_3 = \varepsilon_3$. **SGS still returns the unoriented path $1 - 2 - 3$.**

Example 3: fork, $1 \leftarrow 2 \rightarrow 3$

$$W = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$



$$\widetilde{y}_2 = \varepsilon_2,$$
$$\widetilde{y}_1 = \rho\varepsilon_2 + \varepsilon_1,$$
$$\widetilde{y}_3 = \rho\varepsilon_2 + \varepsilon_3.$$

$$\Sigma_{13} = \rho^2\sigma^2, \quad \Sigma_{12} = \Sigma_{23} = \rho\sigma^2, \quad \Sigma_{22} = \sigma^2.$$

$$\mathrm{Cov}(y_1, y_3 \mid y_2, x) = \Sigma_{13} - \frac{\Sigma_{12}\Sigma_{23}}{\Sigma_{22}} = \rho^2\sigma^2 - \frac{(\rho\sigma^2)^2}{\sigma^2} = 0.$$

$\widetilde{y}_1$ and $\widetilde{y}_3$ share ε_2, so they are marginally dependent. Given $\widetilde{y}_2 = \varepsilon_2$, the rests ε_1 and ε_3 are independent. **Again SGS returns the unoriented path $1 - 2 - 3$.**

Example 4: collider, $1 \to 2 \leftarrow 3$

$$W = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$



$$\widetilde{y}_1 = \varepsilon_1,$$
$$\widetilde{y}_3 = \varepsilon_3,$$
$$\widetilde{y}_2 = \rho\varepsilon_1 + \varepsilon_2 + \rho\varepsilon_3.$$

$$\Sigma_{13} = 0, \quad \Sigma_{12} = \Sigma_{23} = \rho\sigma^2, \quad \Sigma_{22} = (1 + 2\rho^2)\sigma^2.$$

$\widetilde{y}_1 = \varepsilon_1$ and $\widetilde{y}_3 = \varepsilon_3$ share no disturbance, so $\Sigma_{13} = 0$.

SGS removes $1 - 3$ *without conditioning on 2*. This is the key collider signature.

Example 4: conditioning on a collider creates dependence

With $\Sigma_{13} = 0$, $\Sigma_{12} = \Sigma_{23} = \rho\sigma^2$, $\Sigma_{22} = (1 + 2\rho^2)\sigma^2$:

$$\begin{aligned} \mathrm{Cov}(y_1, y_3 \mid y_2, x) &= \Sigma_{13} - \frac{\Sigma_{12}\Sigma_{23}}{\Sigma_{22}} && \text{conditional covariance formula} \\ &= 0 - \frac{(\rho\sigma^2)(\rho\sigma^2)}{(1 + 2\rho^2)\sigma^2} && \text{insert the four entries} \\ &= \boxed{-\frac{\rho^2\sigma^2}{1 + 2\rho^2} \neq 0}. \end{aligned}$$

Why the sign is negative

$\widetilde{y}_2 = \rho\varepsilon_1 + \varepsilon_2 + \rho\varepsilon_3$. Hold $\widetilde{y}_2$ fixed. If $\widetilde{y}_1 = \varepsilon_1$ is large, then $\varepsilon_2 + \rho\varepsilon_3$ must be smaller to keep $\widetilde{y}_2$ fixed, so $\widetilde{y}_3 = \varepsilon_3$ tends to be smaller.

Separating set for 1 and 3: $\varnothing$, which does not contain node 2. Hence SGS orients the collider $1 \to 2 \leftarrow 3$.

Example 5: reciprocal spatial feedback is not a DAG

$$W = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix}, \qquad 0 < \rho < 1.$$



- $w_{13} = w_{31} = 0$: no direct $1 \leftrightarrow 3$ weight.
- W contains two patterns at once:
  fork part $1 \leftarrow 2 \rightarrow 3$ ($w_{12} = w_{32} = 1$),
  collider part $1 \rightarrow 2 \leftarrow 3$
  ($w_{21} = w_{23} = \frac{1}{2}$).
- The system has directed cycles; it is *not* a DAG-SEM on these nodes.

Example 5: even $1-3$ survives the SGS tests

From $\Omega = \frac{1}{\sigma^2}(I-\rho W)^\top (I-\rho W)$,

$$\Omega = \frac{1}{\sigma^2}\begin{pmatrix} 1+\rho^2/4 & -3\rho/2 & \rho^2/4 \\ -3\rho/2 & 1+2\rho^2 & -3\rho/2 \\ \rho^2/4 & -3\rho/2 & 1+\rho^2/4 \end{pmatrix}.$$

Conditional test fails (collider part):

$$\Omega_{13} = \frac{\rho^2 w_{21} w_{23}}{\sigma^2} = \frac{\rho^2}{4\sigma^2} \neq 0.$$

Location 2's equation contains both y_1 and y_3.

Marginal test fails (fork part):

ε_2 enters both $\widetilde{y}_1$ and $\widetilde{y}_3$, so $\Sigma_{13} \neq 0$. All pairwise marginal covariances are nonzero for $0 < \rho < 1$.

SGS outcome

Neither test removes $1-3$: a complete three-node skeleton (triangle), even though W has no direct $1-3$ link.

Summary: original versus exactly field-filtered SGS

SDM input pattern	SGS on original y	SGS on filtered z
Forward chain	Unoriented path $1 - 2 - 3$	No edges
Reverse chain	Unoriented path $1 - 2 - 3$	No edges
Fork	Unoriented path $1 - 2 - 3$	No edges
Collider	$1 \rightarrow 2 \leftarrow 3$	No edges
Reciprocal path	Complete triangle	No edges

The empty filtered skeleton follows from independent Gaussian disturbances and exact realized-field subtraction.

Six tests for all five original-outcome examples

	Marginal			Condition on third		
Case	$1,2$	$1,3$	$2,3$	$1,2 \mid 3$	$1,3 \mid 2$	$2,3 \mid 1$
Forward chain	D	D	D	D	I	D
Reverse chain	D	D	D	D	I	D
Fork	D	D	D	D	I	D
Collider	D	I	D	D	D	D
Reciprocal path	D	D	D	D	D	D

Reading the table

I: independent. **D**: dependent. All tests condition on fixed x.
Marginal columns: zeros of Σ. Conditional columns: zeros of Ω.
For the exactly filtered z, all six entries are **I**.

Realized field versus expected field

Exact (realized) field removal: $\boxed{y - f = \beta x + \varepsilon, \qquad \mathrm{Cov}(y - f \mid x) = \sigma^2 I.}$

Expected field removal: once x is fixed, $\mathrm{E}[f \mid x]$ is a fixed vector. Subtracting a fixed vector moves only the mean: $\boxed{\mathrm{Cov}(y - \mathrm{E}[f \mid x] \mid x) = \mathrm{Cov}(y \mid x) = \Sigma.}$

Explicitly:

$$
\begin{aligned}
y - \mathrm{E}[f \mid x] &= \beta x + (f - \mathrm{E}[f \mid x]) + \varepsilon && y = \beta x + f + \varepsilon \\
&= \beta x + \rho W \widetilde{y} + \varepsilon && f - \mathrm{E}[f \mid x] = \rho W (y - \mathrm{E}[y \mid x]) = \rho W \widetilde{y} \\
&= \beta x + \widetilde{y} && \varepsilon = (I - \rho W)\widetilde{y} = \widetilde{y} - \rho W \widetilde{y}
\end{aligned}
$$

Different operations, different SGS results

The realized field contains the neighbours' actual disturbances, so subtracting it removes them. The expected field contains only their averages, so the mixing $\widetilde{y} = (I - \rho W)^{-1}\varepsilon$ stays, with its original conditional-independence structure.

Why the clean result need not hold in practice

1. **Correlated structural errors:** $\mathrm{Cov}(z \mid x) = \mathrm{Cov}(\varepsilon \mid x)$ need not be diagonal.
2. **Random covariates with unconditional SGS:** $\mathrm{Cov}(z_i, z_j) = \beta^2 \mathrm{Cov}(x_i, x_j)$ if $x \perp\!\!\!\perp \varepsilon$ and errors are mutually independent.
3. **Estimated filters:** insert $\widehat{\rho}, \widehat{\theta}$ in place of ρ, θ:

$$\widehat{z} = y - \widehat{\rho} W y - \widehat{\theta} W x = \beta x + \varepsilon + (\rho - \widehat{\rho}) W y + (\theta - \widehat{\theta}) W x.$$

   The term $(\rho - \widehat{\rho}) W y$ puts some of the neighbours' disturbances back.
4. **Reciprocal dependence:** a simultaneous SDM does not supply the same causal DAG as an acyclic SEM.

A defensible small-network empirical design

1. Fix the three locations, the same covariate, and a stable repeated-sampling design.
2. Specify the directed/reciprocal W and row-normalization convention.
3. Separate the ideal known-parameter filter from the estimated filter.
4. Report all six independence checks, not only the final graph.
5. Compare skeletons and compelled collider arrows separately.
6. Diagnose sampling error, model assumptions, and faithfulness before causal claims.

Sampling warning

One cross-section of three locations is one realization of (y_1, y_2, y_3), not a sample large enough for SGS covariance tests.

Conclusion

1. Exact removal of the *realized* SDM field gives $z = \beta x + \varepsilon$.
2. With fixed x and independent Gaussian errors, the filtered values share no disturbance, so population SGS on z returns an **empty skeleton**.
3. SGS on the original y can return an unoriented path, a compelled collider, or a triangle (fork part and collider part together).
4. Chain and fork orientations are **not** distinguished by observational conditional independence.
5. A sparser filtered graph is not automatically a more accurate causal graph.

Filtering changes the statistical question SGS answers.

Selected references

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Appendix: chain covariance matrix

For $1 \to 2 \to 3$, using $\widetilde{y}_1 = \varepsilon_1$, $\widetilde{y}_2 = \rho\varepsilon_1 + \varepsilon_2$, and $\widetilde{y}_3 = \rho^2\varepsilon_1 + \rho\varepsilon_2 + \varepsilon_3$:

$$\Sigma = \sigma^2 \begin{pmatrix} 1 & \rho & \rho^2 \\ \rho & 1+\rho^2 & \rho+\rho^3 \\ \rho^2 & \rho+\rho^3 & 1+\rho^2+\rho^4 \end{pmatrix}.$$

$\Omega_{12} = \Omega_{23} = -\rho/\sigma^2 \neq 0$, but $\Omega_{13} = 0$.

The conditional SGS checks preserve adjacent pairs and remove 1–3.

Appendix: collider covariance and precision

For $1 \to 2 \leftarrow 3$:

$$\Sigma = \sigma^2 \begin{pmatrix} 1 & \rho & 0 \\ \rho & 1+2\rho^2 & \rho \\ 0 & \rho & 1 \end{pmatrix},$$

$$\Omega = \frac{1}{\sigma^2} \begin{pmatrix} 1+\rho^2 & -\rho & \rho^2 \\ -\rho & 1 & -\rho \\ \rho^2 & -\rho & 1+\rho^2 \end{pmatrix}.$$

The zero Σ_{13} removes $1-3$ marginally; nonzero Ω_{13} means conditioning on 2 induces dependence.

Appendix: why Ω_{ij} tests conditional independence

The inverse of a matrix is its cofactor matrix, transposed, divided by its determinant. For the pair $(1,3)$ given 2: delete row 3 and column 1 of Σ.

$$\begin{aligned}
\Omega_{13} &= \frac{1}{\det \Sigma} \det \begin{pmatrix} \Sigma_{12} & \Sigma_{13} \\ \Sigma_{22} & \Sigma_{23} \end{pmatrix} && \text{cofactor formula for the inverse} \\
&= \frac{\Sigma_{12}\Sigma_{23} - \Sigma_{13}\Sigma_{22}}{\det \Sigma} && 2 \times 2 \text{ determinant} \\
&= -\frac{\Sigma_{22}}{\det \Sigma} \left(\Sigma_{13} - \frac{\Sigma_{12}\Sigma_{23}}{\Sigma_{22}} \right) && \text{factor out } -\Sigma_{22} \\
&= \boxed{-\frac{\Sigma_{22}}{\det \Sigma} \operatorname{Cov}(y_1, y_3 \mid y_2, x)} && \text{conditional covariance formula}
\end{aligned}$$

Σ is positive definite, so $\Sigma_{22} > 0$ and $\det \Sigma > 0$. Hence $\Omega_{13} = 0 \iff \operatorname{Cov}(y_1, y_3 \mid y_2, x) = 0$. The pairs $(1,2)$ and $(2,3)$ work the same way.

Ω_{ij} and the conditional covariance have opposite signs. Collider check: $\Omega_{13} = \rho^2/\sigma^2 > 0$ while $\operatorname{Cov}(y_1, y_3 \mid y_2, x) < 0$.

Appendix: reciprocal feedback algebra

The centered simultaneous equations are

$$\widetilde{y}_1 = \rho\widetilde{y}_2 + \varepsilon_1, \quad \widetilde{y}_2 = \frac{\rho}{2}(\widetilde{y}_1 + \widetilde{y}_3) + \varepsilon_2, \quad \widetilde{y}_3 = \rho\widetilde{y}_2 + \varepsilon_3.$$

$$\widetilde{y}_2 = \frac{\rho}{2}\left[(\rho\widetilde{y}_2 + \varepsilon_1) + (\rho\widetilde{y}_2 + \varepsilon_3)\right] + \varepsilon_2 \qquad \text{insert } \widetilde{y}_1 \text{ and } \widetilde{y}_3 \text{ into the middle equation}$$

$$(1-\rho^2)\widetilde{y}_2 = \frac{\rho}{2}\varepsilon_1 + \varepsilon_2 + \frac{\rho}{2}\varepsilon_3 \qquad \text{move } \rho^2\widetilde{y}_2 \text{ to the left}$$

$$\widetilde{y}_2 = \frac{\rho\varepsilon_1/2 + \varepsilon_2 + \rho\varepsilon_3/2}{1-\rho^2} \qquad \text{divide by } 1-\rho^2 > 0$$

$$\widetilde{y}_1 = \rho\widetilde{y}_2 + \varepsilon_1 = \frac{(1-\rho^2/2)\varepsilon_1 + \rho\varepsilon_2 + (\rho^2/2)\varepsilon_3}{1-\rho^2} \qquad \text{insert } \widetilde{y}_2\text{; common denominator}$$

$$\widetilde{y}_3 = \frac{(\rho^2/2)\varepsilon_1 + \rho\varepsilon_2 + (1-\rho^2/2)\varepsilon_3}{1-\rho^2} \qquad \text{same steps, by symmetry}$$

All three contain ε_2 (the fork part), so every marginal covariance is nonzero.

Appendix: feedback conditional covariance

Shared-disturbance rule on the three mixes above, with $d = 1 - \rho^2$:

$$\Sigma_{13} = \frac{\sigma^2(2\rho^2 - \rho^4/2)}{d^2}, \qquad \Sigma_{12} = \Sigma_{23} = \frac{3\rho\sigma^2}{2d^2}, \qquad \Sigma_{22} = \frac{(1 + \rho^2/2)\sigma^2}{d^2}.$$

$$\begin{aligned}
\mathrm{Cov}(y_1, y_3 \mid y_2, x) &= \Sigma_{13} - \frac{\Sigma_{12}\Sigma_{23}}{\Sigma_{22}} && \text{conditional covariance formula} \\
&= \frac{\sigma^2}{d^2}\left[2\rho^2 - \frac{\rho^4}{2} - \frac{9\rho^2/4}{1 + \rho^2/2}\right] && \text{insert; the factor } \sigma^2/d^2 \text{ is common} \\
&= \frac{\sigma^2}{d^2} \cdot \frac{-\frac{\rho^2}{4} + \frac{\rho^4}{2} - \frac{\rho^6}{4}}{1 + \rho^2/2} && \text{common denominator; expand} \\
&= \frac{\sigma^2}{d^2} \cdot \frac{-\frac{\rho^2}{4}(1 - \rho^2)^2}{1 + \rho^2/2} && 1 - 2\rho^2 + \rho^4 = (1 - \rho^2)^2 = d^2 \\
&= \boxed{-\frac{\rho^2\sigma^2}{4(1 + \rho^2/2)} \neq 0} && \text{cancel } d^2
\end{aligned}$$

Appendix: why a three-node DAG gives a finite inverse

An acyclic three-node spatial-weights matrix with zero diagonal has $W^3 = 0$: a directed path with three arrows would need four distinct nodes.

$$
\begin{aligned}
&(I - \rho W)(I + \rho W + \rho^2 W^2) \\
&\quad = I + \rho W + \rho^2 W^2 - \rho W - \rho^2 W^2 - \rho^3 W^3 && \text{multiply out} \\
&\quad = I - \rho^3 W^3 = I && W^3 = 0
\end{aligned}
$$

Thus the reduced form in the acyclic examples can be written without an infinite series:

$$\boxed{(I - \rho W)^{-1} = I + \rho W + \rho^2 W^2.}$$

The W^2 term carries dependence along two-arrow paths. This matrix is the mixing table of the acyclic examples.